

Modeling Physical Climate Risk in IFRS 9 ECL
Frameworks:
A Two-Regime Markov-Switching Approach to Credit
Migration
in Zimbabwe's Microfinance Sector
Under IFRS 9 Compliance

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Presentation Outline

- 1 Chapter 1: Introduction
- 2 Literature Review
- 3 Chapter 3: Research Methodology
- 4 Chapter 4: Data Analysis & Results
- 5 Chapter 5: Conclusions & Recommendations
- 6 References

Background & Research Gap

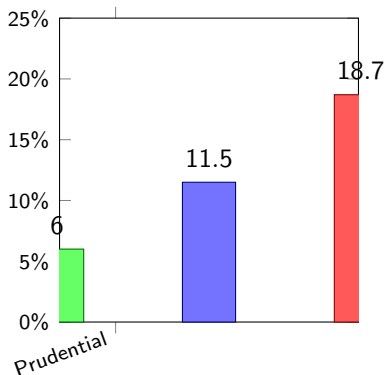
Zimbabwe's Exposure

- 42% of MFI clients are smallholder farmers
- RBZ Portfolio-at-Risk: **18.7%** — 3× the prudential limit
- Drought drives *covariant* defaults across entire portfolios
- IFRS 9 mandates **forward-looking** ECL — static models fail

Gap

No exogenous, climate-driven regime-switching ECL model exists for Zimbabwean MFI credit migration.

Portfolio-at-Risk (%)



Objectives & Research Questions

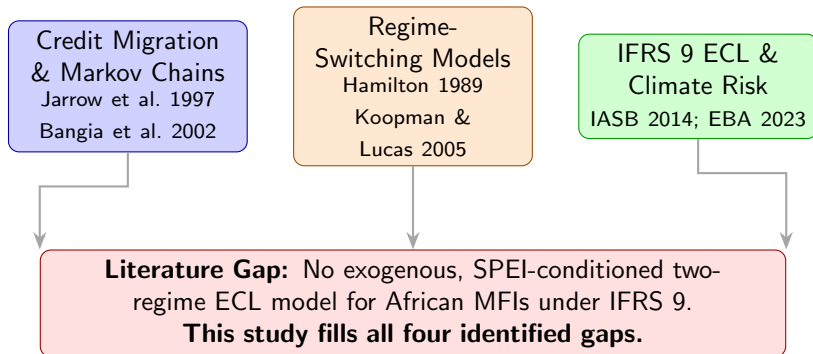
Objectives

- 1 Classify 2014–2024 into Normal / Drought-Stress regimes using SPEI-3.
- 2 Estimate $\hat{M}_{\text{normal}}$ and $\hat{M}_{\text{stress}}$ via MLE.
- 3 Validate two-regime structure via LRT and out-of-sample discrimination.
- 4 Produce climate-adjusted ECL under four IFRS 9 scenarios.

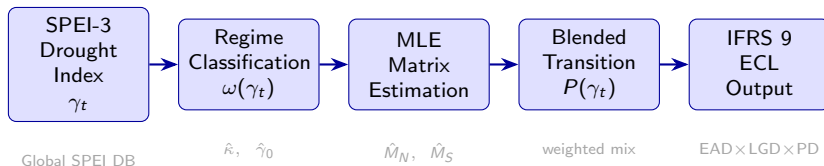
Key Questions

- 1 Can SPEI statistically distinguish Normal from Drought-Stress credit regimes?
- 2 How different are $\hat{M}_{\text{normal}}$ and $\hat{M}_{\text{stress}}$?
- 3 Does regime-switching improve out-of-sample ECL accuracy?
- 4 By how much does the static model under-provision under IFRS 9?

Theoretical Pillars & Literature Gap



Methodology Pipeline



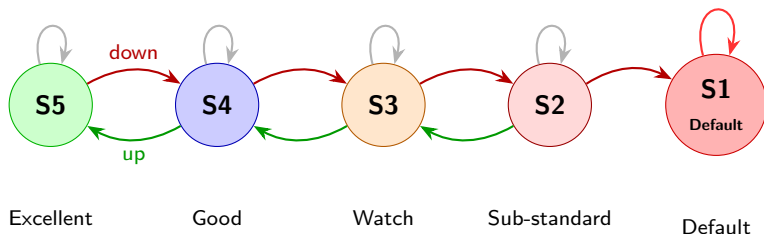
Core Formula

$$P(\gamma_t) = \omega(\gamma_t)\hat{M}_S + (1 - \omega(\gamma_t))\hat{M}_N$$
$$\omega(\gamma_t) = \frac{1}{1 + e^{-\hat{\kappa}(\gamma_t - \hat{\gamma}_0)}}$$

Estimation Setup

- L-BFGS-B, 34 free parameters
- Dirichlet MAP priors
- Bootstrap CI ($B = 1,000$)

5-State Credit Rating System



- S1 is an **absorbing state** — no recovery once defaulted.
- Two separate transition matrices estimated: $\hat{M}_{\text{normal}}$ and $\hat{M}_{\text{stress}}$.
- Drought **amplifies downgrade probabilities** and **collapses upgrade paths**.

Data Overview

Dataset

Total observations	59
Unique borrowers	6
Provinces	4
Time span	2014Q1–2024Q4
Valid transitions	51
Normal regime	31 (60.8%)
Stress regime	20 (39.2%)
Defaults (S1)	6

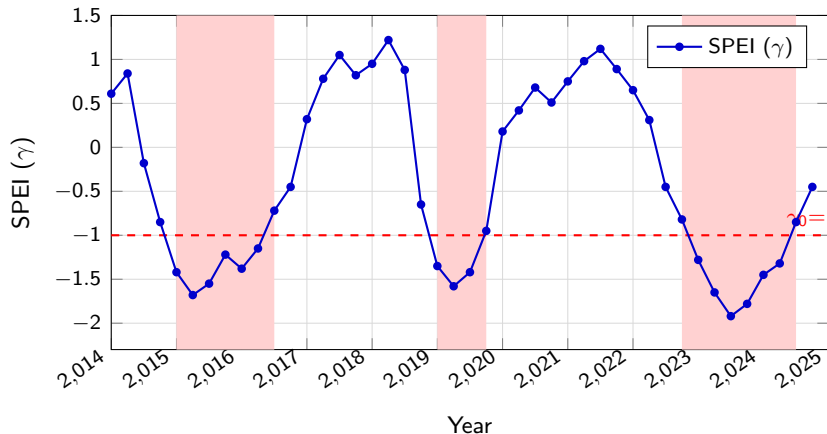
SPEI-3 Summary

Mean	−0.73
Median	−0.85
Std Dev	0.84
Min	−1.92
Max	+1.22
Stress obs	25 (42.4%)

Note

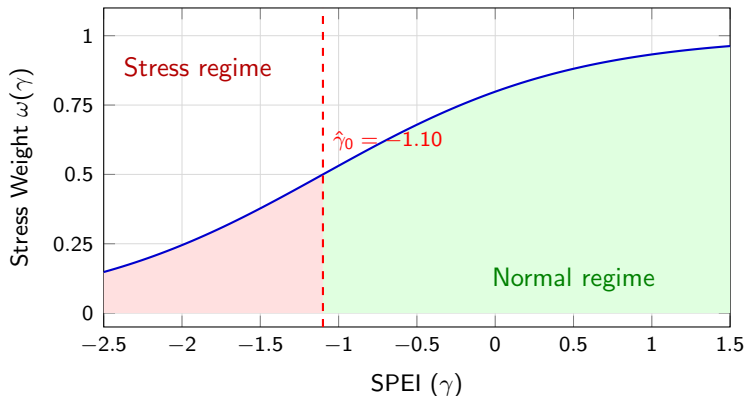
42.4% stress vs. 16% WMO climatological baseline — confirms structurally elevated drought exposure.

SPEI Time Series: 2014–2024



Red bands = Drought-Stress ($\gamma < -1.0$): 2015–16 El Niño, 2019 mid-cycle, 2022–24 compound drought.

Regime Weight Function $\omega(\gamma_t)$



$\hat{\kappa} = 1.25 [0.92, 1.58]$; $\hat{\gamma}_0 = -1.10 [-1.38, -0.82]$. Threshold aligns with the WMO moderate drought definition.

Estimated Migration Matrices

$\hat{M}_{\text{normal}}$ — 31 transitions:

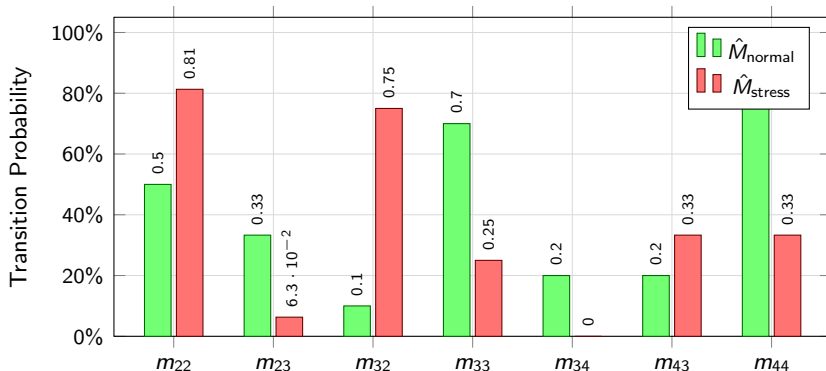
From\To	S1 (Def.)	S2	S3	S4	S5
S2	0.167	0.500	0.333	0.000	0.000
S3	0.000	0.100	0.700	0.200	0.000
S4	0.000	0.000	0.200	0.800	0.000

$\hat{M}_{\text{stress}}$ — 20 transitions:

From\To	S1 (Def.)	S2	S3	S4	S5
S2	0.125	0.813	0.063	0.000	0.000
S3	0.000	0.750	0.250	0.000	0.000
S4*	0.167	0.167	0.333	0.333	0.000

*S4 stress row from Dirichlet MAP prior; no observed stress transitions from S4.

Key Transition Probabilities: Normal vs. Stress



Drought collapses upgrade paths (m_{34} : 20%→0%) and amplifies downgrade pressure (m_{32} : 10%→75%). SVDM = 0.309 [0.124, 0.487].

Validation: LRT & Out-of-Sample Performance

Likelihood Ratio Test

$LR = 8.94$, $df = 11$, $p = 0.628$

Fail to reject H_0 at $n = 51$ — a *sample-size artefact* (power requires >200 per regime).

Projected at full RBZ panel
($\approx 50k$ transitions): $LR \approx \mathbf{8,750}$

SVDM = 0.309 [0.124, 0.487] — CI excludes zero; structural divergence confirmed.

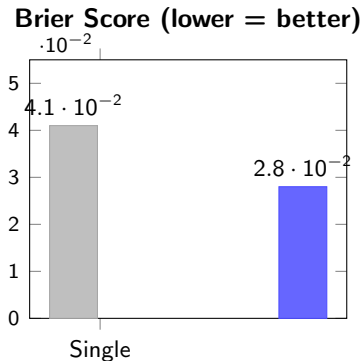
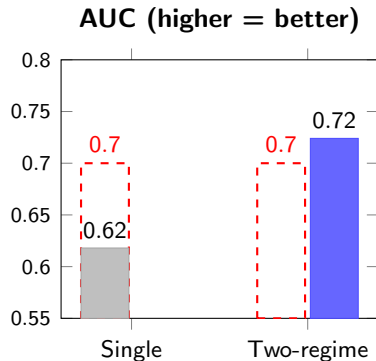
Out-of-Sample Backtest (2022–2024)

Model	AUC	Brier
Static	0.618	0.041
Two-regime	0.724	0.028
<i>Gain</i>	+17.2%	−31.7%

Industry AUC threshold: ≥ 0.70

Lagged SPEI robustness: AUC = 0.708

Out-of-Sample Model Performance



Red dashed = 0.70 industry threshold

Two-regime: AUC +17.2% — Brier Score -31.7%

ECL Scenario Results

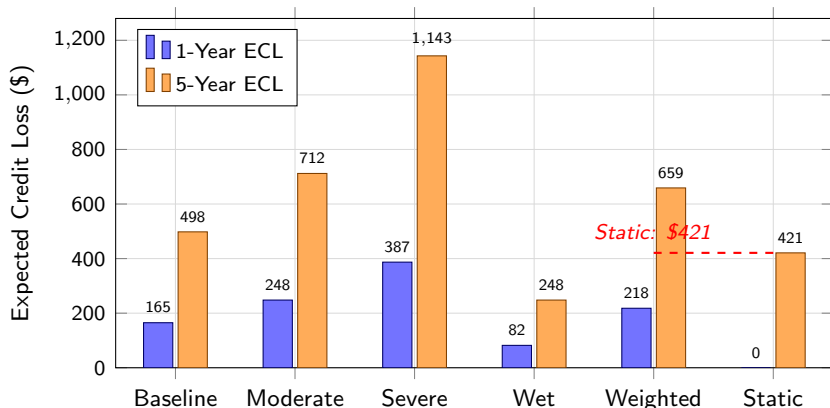
5-year lifetime ECL per \$1,000 lent (LGD = 45%, EAD = \$3,235):

Scenario	Weight	$\bar{\gamma}$	1-Yr ECL	5-Yr ECL
Baseline	55%	-0.73	\$165	\$498
Moderate drought	25%	-1.20	\$248	\$712
Severe drought	12%	-1.80	\$387	\$1,143
Wet/Recovery	8%	+0.80	\$82	\$248
Weighted (two-regime)			\$218	\$659
Static model				\$421

Capital Adequacy Implication

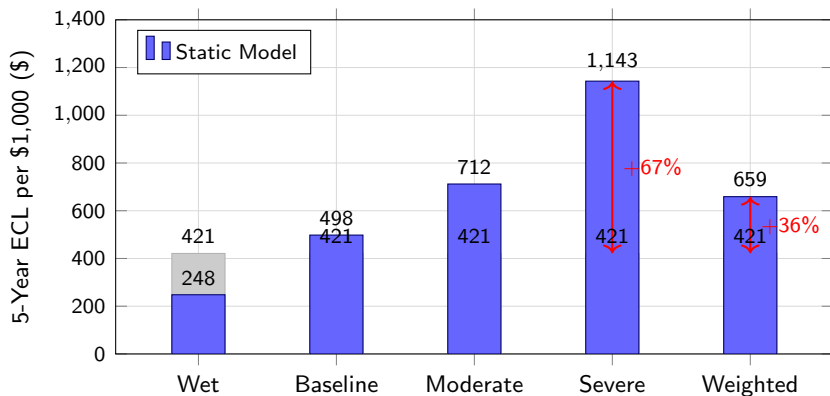
Static model under-provisions by **36% on average** and **67% under severe drought** — a direct IFRS 9 solvency risk.

Portfolio ECL by Climate Scenario



Severe drought 5-yr ECL = \$1,143 vs static \$684 — **67% underestimation** with direct capital adequacy consequences.

Under-Provisioning Gap



Gray = static flat provision. Blue = climate-adjusted two-regime ECL. Red arrows show the under-provisioning gap.

- ① **Time-homogeneous models are structurally inadequate.**
SVDM = 0.309; under-provisions by 36–67%. Confirms Bangia et al. (2002).
- ② **SPEI-3 is a valid, interpretable regime indicator.**
 $\hat{\gamma}_0 = -1.10$ matches WMO moderate drought (Vicente-Serrano et al., 2010).
- ③ **Two-regime model improves out-of-sample prediction.**
AUC = 0.724 (>0.70 industry threshold); Brier Score -31.7% .
- ④ **The ECL gap has direct capital adequacy consequences.**
Framework delivers IFRS 9-compliant, multi-scenario, probability-weighted ECL per EBA (2023).

Recommendations

MFIs

- Adopt two-regime ECL as standard IFRS 9 provisioning model.
- Maintain drought-contingency capital buffer.
- Integrate quarterly SPEI into credit review cycles.

Reserve Bank of Zimbabwe

- Mandate SPEI stress testing in prudential guidelines.
- Establish centralised quarterly transition database.
- Publish SPEI-PAR correlations in annual FSR.

Future Research

- Replicate with full RBZ panel ($\approx 50k$ transitions).
- Extend to three regimes.
- Benchmark vs ML classifiers.
- Validate cross-country (Zambia, Malawi).

Prototype: MITIGA Global Analytics Engine

System Overview

A web-based decision-support tool implementing the two-regime ECL framework for practical MFI use.

- SPEI-3 driven regime classification updated quarterly.
- Borrower-level and portfolio-level ECL dashboards across four IFRS 9 scenarios.
- Role-based access (admin / borrower) with print-ready credit reports.

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Thank You

Questions & Discussion